Image Super-Resolution Using SRCNN

UE24CS352A Machine Learning Mini-Project Report | Rohit Chaki (PES1UG24AM228), Pulkit Handa (PES1UG24AM208)

1. Problem Statement & Context

Single Image Super-Resolution (SISR) aims to reconstruct crisp, high-resolution (HR) images from low-resolution (LR) inputs. Traditional math-based interpolation methods like bilinear or bicubic upscaling tend to produce blurry edges because they only smooth neighboring pixels without understanding underlying image structures. In this project, we re-implemented the Super-Resolution Convolutional Neural Network (SRCNN) based on Dong et al. and the CS229 Stanford benchmark (Garber et al., 2020) to evaluate how a simple three-layer network learns feature mappings directly from data to sharpen images.

2. Dataset & Preprocessing

We used 100 high-resolution images from the public DIV2K dataset, following a strict 60/20/20 train/validation/test split to prevent data leakage:

• **Crop & Target Prep:** Each image was center-cropped to 800x800 and scaled down to 224x224 pixels to create the high-resolution target (HR).

• **Downscaling Degradation:** To generate low-resolution inputs (LR), the 224x224 targets were downsized to 112x112 (a 2x reduction) and stretched back up to 224x224 using bicubic interpolation, introducing natural-looking blur.

• **Normalization:** Pixel values were normalized into [0, 1] RGB tensors to ensure smooth gradient updates during training.

3. Network Architecture & Methodology

SRCNN replaces complex manual feature extraction with three simple convolutional steps operating directly on RGB images:

1. Patch Extraction: A 9x9 convolution (64 filters, padding 4) extracts feature patches from the blurred input image.

2. Non-Linear Mapping: A 1x1 convolution (32 filters) non-linearly maps the 64 feature maps into 32 reduced representations, followed by a ReLU activation.

3. Reconstruction: A 5x5 convolution (3 filters, padding 2) combines these feature maps back into a clean 3-channel RGB image.

The network has ~20,000 parameters and was trained end-to-end using Mean Squared Error (MSE) loss and the Adam optimizer (learning rate $\eta = 0.001$).

4. Training Strategy & Hyperparameters

We ran experiments over 100 epochs to find the right training setup:

• **Batch Size Dynamics:** With a batch size of 16, the model updated weights too infrequently (~4 updates/epoch), leading to undertraining and a test PSNR of 26.87 dB (worse than bicubic). Dropping the batch size to 2 increased updates to ~30 per epoch, allowing the model to cross the bicubic threshold around epoch 50 and reach a peak validation PSNR of 27.33 dB.

• **Normalization & Regularization:** Adding batch normalization caused noticeable instability and lower PSNR, while L2 weight decay offered no gains. Standard ReLU without batch norm gave the cleanest results.

5. Results & Analysis

We evaluated all methods on the 20 unseen test images. Table 1 outlines the quantitative scores:

Method / Configuration	Test MSE	Test PSNR (dB)
Nearest Neighbor Baseline	0.00282	26.17 dB
Bilinear Interpolation Baseline	0.00253	26.76 dB
Bicubic Interpolation Baseline	0.00199	27.86 dB
SRCNN (Batch Size = 16, 100 epochs)	0.00244	26.87 dB
SRCNN (Batch Size = 2, 100 epochs - Optimal)	0.00171	**28.48 dB**

Table 1: Test performance across baselines and SRCNN configurations.

> **Key Result:** *Our tuned SRCNN achieved a test PSNR of 28.48 dB, outperforming bicubic interpolation by +0.62 dB and demonstrating that smaller batch sizes were essential for effective parameter updates on a compact dataset.*

6. Discussion & Qualitative Insights

Visually, SRCNN outputs show noticeably sharper edges and cleaner textures compared to bicubic upscaling. However, a few trade-offs were observed:

• **Boundary Artifacts:** Zero-padding at the outer convolutional borders occasionally left subtle edge artifacts.

• **Image Content Variation:** Performance varied by texture type—structured patterns (e.g., lattice grids) gained over +1.0 dB PSNR, whereas fine organic hair and foliage details saw occasional minor drops (-0.8 dB) due to limited dataset diversity.

7. Conclusion & Next Steps

This mini-project demonstrates that even a compact CNN outperforms traditional mathematical upscaling for super-resolution tasks. In future work, we plan to train on the full 800-image DIV2K dataset, experiment with deeper architectures like U-Net or GAN-based perceptual losses, and incorporate SSIM metrics for better perceptual quality assessment.